# Infrastructure Capacity, Risk, and Firm Value: Evidence from U.S. Electricity Tightness (Kacperczyk & Licher)

Dicussed by Johan Hombert (HEC Paris)

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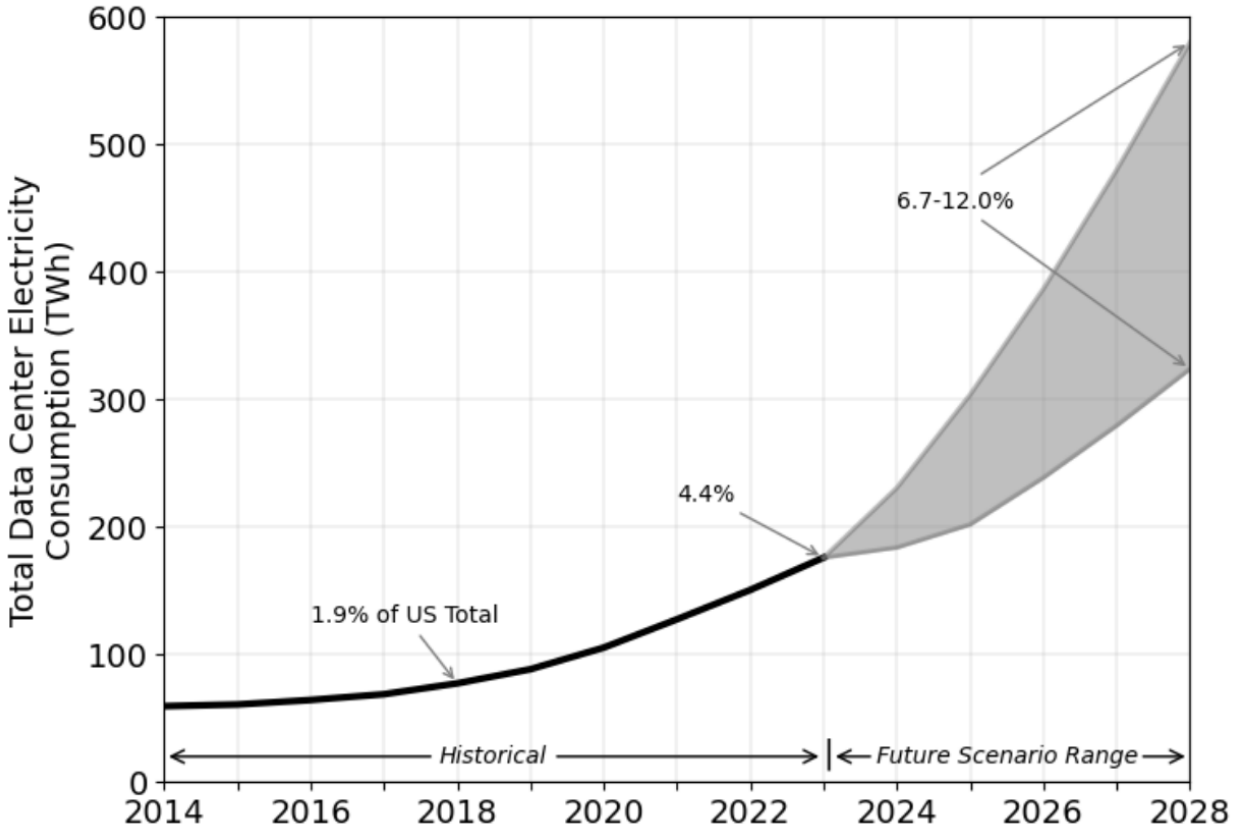
# Summary

Increasing electricity demand (data centers, electrification of transport, manufacturing, and buildings)




$\Rightarrow$ Risk of hitting capacity constraints (generation and grid transmission)

# Summary

Risk of power outage leads to:

(1) Real effects: lower employment, investment, and profit

Electricity is complement with $K$ and $L$. Outage $\Rightarrow K, L \downarrow \Rightarrow Y, \Pi \downarrow$

(2) Priced risk: higher cost of capital

If outage risk is systematic, then realized outage $\Rightarrow$ lower return AND lower agg. production $\Rightarrow$ exposure to outage risk is priced

# Comments

1. Exposure intensity

2. Quantitative interpretation

3. Systematic outage risk

# 1. Exposure Intensity

- Model: Effect of outage risk depends on the firm's electricity intensity ($\delta_E$)
- Empirics: Average effect of (location-specific) outage risk
- Suggestion: Use cross-sectional variation in electricity intensity (e.g., measured at industry level, in the spirit of Table 12 Panels C/D using NAICS 518210 "Data processing, hosting, and related services")
- Exposure may differ for on-demand services (data centers) vs. production that can be halted for a few hours (AI training, manufacturing)
- Hyperscaler data centers significant now/in the future but less in-sample (2005-2024). Off which industries are the effects identified?

# 2. Quantitative Interpretation

- Reverse causality? Higher investment and production $\Rightarrow$ Higher electricity consumption $\Rightarrow$ Higher risk of outage

$\Rightarrow$ Downward bias?

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- Aggregation? Diff-in-diff across counties $\Rightarrow$ Cannot distinguish between activity moving across counties vs. aggregate effect (missing intercept)

$\Rightarrow$ Local effect = upper bound for aggregate effect?

# 3. Systematic Outage Risk

- Why is outage risk priced?
- Model: risk is systematic because there is one location only $\Rightarrow$ outage affects aggregate production
- Data: outages are local (consistent with cross-county identification) $\Rightarrow$ macro effect of local outage is likely small
- Systematic outage risk may stem from uncertainty about longer-term increase in electricity demand and in capacity ramp-up $\Rightarrow$ aggregate risk with heterogeneous exposure across locations

# Conclusion

- Brings model and new evidence on an important topic
- Suggestions to tighten identification and interpretation
- Looking forward to read the next version